

Saksham Shrestha

Data Science Student | Python Developer

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TECHNICAL SKILLS

Programming Languages: Python, SQL Machine Learning & AI: Scikit-Learn, XGBoost, LangChain, FAISS (Vector DB), RAG Architecture, Gemini API, Time-Series Forecasting, Feature Engineering Data Science & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Model Evaluation & Preprocessing Tools & Frameworks: Streamlit, Git, GitHub, GitHub Actions, Joblib, Jupyter Notebook, VS Code Databases & APIs: MySQL, SQLite, REST APIs, STAC API, Web Scraping, Query Optimization

EDUCATION

Bachelor of Science in Computer Science & Information Technology (BSc. CSIT) Amrit Campus, Kathmandu | December 2023 – Present

- Currently pursuing undergraduate degree in Computer Science

Data Science with Python Training Broadway Infosys Nepal, Kathmandu | December 2025 – Present

- Ongoing professional certification in data science and machine learning

NEB 10+2 (Science) Xavier Academy, Kathmandu | 2020 – 2022

- GPA: 3.7 out of 4.0
- Relevant Coursework: Mathematics, Physics

PROFESSIONAL EXPERIENCE

Inventory Associate Buneko Nepal, Kathmandu | July 2023 – August 2023

- Maintained inventory records with 100% accuracy using data management systems
- Performed data entry and verification for stock tracking and management
- Supported operational workflows with attention to detail and accuracy
- Ensured zero deficiencies in inventory logs through systematic verification

PROJECTS

Satellite-Based Crop Health Predictor | Python, XGBoost, Scikit-Learn, Pandas, Streamlit, Planetary Computer API, Joblib

[GitHub](#) | [Live Demo](#)

- Ingested multi-modal geospatial and meteorological data by querying Sentinel-2 imagery via the Planetary Computer STAC API and harvesting daily climate metrics from the Open-Meteo API
- Processed high-resolution satellite bands to calculate cloud-masked scene NDVI and engineered 14-day rolling features including moisture deficit, water satisfaction ratio, and heatwave indicators
- Trained and evaluated an XGBoost Regressor on chronologically split time-series data to forecast plant vigor (NDVI) 14 days in advance using MAE and R^2 metrics
- Developed an interactive Streamlit dashboard featuring baseline vs. forecasted NDVI predictions, actual-vs-predicted trend visual charts, and dynamic historical date filters
- Orchestrated an end-to-end modular pipeline automating raw data fetching, feature creation, model execution, and artifact serialization via Joblib for decoupled application serving

Nutritional Parameter Analysis Engine | Python, Scikit-Learn, Pandas, Joblib, Streamlit

[GitHub](#) | [Live Demo](#)

- Trained and evaluated classification models (Logistic Regression, Decision Trees, Random Forest) on a 7-feature core nutrient vector to predict food healthiness
- Preprocessed raw data with custom string sanitization, missing-value handling, and normalized numeric inputs using StandardScaler
- Built an interactive Streamlit dashboard featuring dynamic serving-size scaling and traffic light visual status indicators
- Integrated hybrid rule-based health profile guardrails (Diabetic, Low Sodium, Keto) to override ML predictions when critical nutrient limits are breached
- Serialized trained model binaries and scaler artifacts using joblib with an automated heuristic fallback system for robust fault tolerance

Banking Support Agent | Python, LangChain, Gemini API, FAISS, Streamlit

[GitHub](#)

- Built a RAG pipeline that embeds banking FAQ documents into a FAISS vector store using Google Generative AI embeddings
- Implemented semantic similarity search to retrieve the most relevant document chunks for any user query
- Designed a modern LCEL chain with prompt templating to generate grounded, hallucination-free responses via Gemini
- Developed a Streamlit chat interface with session state management for persistent conversation history

F1 Race Day Notifier Bot | Python, Pandas, Viber API, GitHub Actions

[GitHub](#)

- Fetched live Formula 1 session data from the OpenF1 REST API and parsed it into a pandas DataFrame
- Applied datetime filtering logic to detect next-day races and extract location and time details
- Delivered automated race alerts to Viber messenger via the Viber Bot API
- Deployed on GitHub Actions to run on a daily schedule with no manual intervention or local machine required

Stock Market Data Collector & Viewer | Python, MySQL

GitHub

- Developed Python scripts to fetch and process daily stock market data from APIs
- Performed data cleaning, validation, and normalization to ensure data integrity
- Stored structured financial data in a MySQL database and executed SQL queries for analysis
- Automated CSV exports and implemented a CLI-based search for stock lookup

CERTIFICATIONS

- SQL (Basic) Certification | HackerRank | 2026
- Bronze Honor Recipient | International Youth Math Challenge (IYMC) | 2023
Ranked in top 15% globally for problem-solving and mathematical reasoning

ACHIEVEMENTS

Xavier Academy Science Exhibition Organizer | 2022

- Successfully organized and coordinated annual science exhibition
- Managed team collaboration and event logistics

Robotics Club Member | Xavier Academy | 2022

- Participated in robotics workshops and team projects
- Developed problem-solving and technical skills through hands-on activities

VOLUNTEER EXPERIENCE

Member | Nepali Pariwar | 2022 – Present

- Raised funds and donated supplies to Heartbeat Foundation Nepal
- Organized weekly community cleaning campaigns for environmental sustainability
- Coordinated donations to local institutions for social welfare

Volunteer | Food for Life Nepal | 2019

- Participated in fundraising campaign for children's welfare and education
- Assisted in event management and community outreach activities

Summary

Computer Science undergraduate with proven experience building end-to-end machine learning models, RAG architecture, and automated data pipelines. Demonstrated track record in predictive forecasting (XGBoost, Scikit-Learn), geospatial and time-series feature engineering, and LLM integration using LangChain, Google Gemini API, and FAISS vector stores. Proficient in delivering interactive Streamlit dashboards and orchestrating automated API workflows with GitHub Actions and SQL databases. Seeking an AI/ML internship to apply expertise in predictive modeling and generative AI to scalable, real-world intelligent systems.